Trader Performance vs Market Sentiment

Behavioral Analysis & Strategy Framework

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Colab_link: “https://colab.research.google.com/drive/1jj1o6bIcV3IkRz0JfkBumxC-1hBGFoKG?usp=sharing”

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1. Executive Summary

This report analyzes the relationship between **Bitcoin market sentiment** and **trader behavior** using the Fear & Greed Index and historical trade-level data.
The objective is to uncover **hidden behavioral patterns**, evaluate **risk-adjusted performance**, and derive **actionable trading strategies**.
Key findings reveal that trader performance is driven more by **behavioral discipline and risk control** than by raw profitability. Market sentiment significantly influences **position sizing, leverage usage, asset selection**, and **trade quality**, creating exploitable patterns for smarter trading decisions.

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2. Data Overview

Datasets Used

- **Bitcoin Market Sentiment Dataset**
    - Daily sentiment classification: *Fear, Greed, Neutral*
- **Hyperliquid Historical Trading Data**
    - Trade-level data including size, PnL, leverage proxy, asset, and timestamps

Scope

- 200K+ trades analyzed
- Multiple sentiment regimes
- Trader-level and asset-level aggregation

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3. Methodology

The analysis followed a structured pipeline:

1. Data cleaning and temporal alignment
2. Feature engineering (sentiment regimes, leverage proxy, trade quality metrics)
3. Descriptive and advanced cross-variable analysis
4. Time-series and regime shift analysis
5. Behavioral pattern discovery
6. Strategy formulation

Statistical testing and clustering techniques were used to validate non-random behavioral patterns.

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4. Key Findings & Pattern Discoveries

4.1 Revenge Trading Behavior (Position Sizing Anomalies)

- Traders increase position size by **42.2% after losses**
- Behavior is **statistically significant ($p < 0.001$)**
- Effect is amplified during **Fear regimes**
- A subset of traders consistently exhibits revenge trading patterns

Implication:
Loss aversion leads to compounding risk and deeper drawdowns instead of recovery.

4.2 Early Indicators of Regime Shifts

Behavioral changes **precede sentiment regime transitions**:

- **PnL volatility spikes by ~480%**
- **Position sizes drop by ~20%**
- **Win rates deteriorate**

Implication:
Volatility expansion combined with risk reduction acts as a **leading indicator** of regime change, allowing proactive strategy adjustment.

4.3 Trader Style Clustering

Clustering reveals **four distinct trader archetypes**:

- **Elite Performers (~9%)**
  High win rate, controlled exposure, consistent profitability
- **High-Risk Gamblers (~81%)**
  Extreme leverage, volatile outcomes, low survivability
- **High-Frequency Traders (~9%)**
  Profit via execution speed and volume
- **Conservative Players (~0%)**
  Rare in dataset, capital preservation focused

Implication:
Consistent success is linked to **discipline**, not aggression.

4.4 Risk-Adjusted Performance vs Absolute PnL

- Only **1 out of 10** traders overlap between:
  - Top absolute PnL
  - Top risk-adjusted returns
- Many profitable traders achieve returns through **excessive volatility**

Best Risk-Adjusted Regime: Neutral

Worst Risk-Adjusted Regime: Fear

Implication:
Raw profits mask fragility. Risk-adjusted metrics better represent true skill.

4.5 Asset-Specific Regime Performance

Clear **coin–sentiment dependencies** emerge:

- **Greed Regime:**
  @107, HYPE deliver outsized returns
- **Fear Regime:**
  SOL, ETH show stronger resilience
- **Neutral Regime:**
  BTC remains relatively stable

Implication:
Sentiment-aware asset rotation provides regime-specific alpha.

4.6 Trade Quality Score & Performance

- Higher trade-quality deciles show:
  - Higher win rates (up to ~64%)
  - More stable PnL
- Low-quality trades dilute performance despite higher frequency

Implication:
Trade selection quality outweighs trade quantity.

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5. Actionable Trading Framework

Based on all findings, a **systematic decision model** is proposed:

Core Rules

- Reduce position size by **40% during Fear**
- Cap leverage:
  - **10× in Fear**
  - **15× in Greed**
- Never increase size after a losing trade
- Trade only regime-favorable assets
- Monitor volatility spikes as regime change warnings
- Align strategy with trader archetype strengths
- Prioritize high-quality trades over frequency

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6. Strategic Takeaways

- Market sentiment strongly shapes trader behavior and risk outcomes
- Emotional responses (loss aversion, overconfidence) drive suboptimal decisions
- Elite performance is defined by **discipline, patience, and risk control**
- Sentiment-aware strategies materially improve capital efficiency

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7. Conclusion

This analysis demonstrates that **behavioral signals embedded in trading data provide actionable insights** beyond traditional performance metrics.
By integrating sentiment, risk-adjusted evaluation, and behavioral controls, traders can significantly reduce drawdowns and improve long-term performance.
Smarter trading is not about trading more — it is about trading better.